

PROJECT BLUEPRINT

1. Project Overview

Project Title - Sentix Pulse: Real-Time Sentiment Analyser for Product Reviews

One-line Problem Statement - Traditional rating systems (like star ratings) fail to capture nuanced technical feedback and hidden product performance issues buried within textual reviews.

Objective - To design an AI-powered sentiment intelligence system that:

- Converts raw product reviews into actionable business intelligence
- Generates a Verified Buy Score for consumers
- Extracts Pain-Point Clusters for corporate product teams

Target Users

- Tech-savvy consumers (decision-making support)
- Product Managers & Business Analysts (market intelligence)

Expected Outcome- A dual-portal web intelligence platform that transforms unstructured reviews into:

- Verified Buy Confidence Index
- Competitive product insights
- Visual dashboards + strategic NLP-generated summaries

2. Problem Statement

Existing Problem - Modern review platforms suffer from:

- Fake / biased reviews
- Over-reliance on star ratings
- Hidden technical failures inside textual feedback
- Lack of structured attribute-level analysis (battery, performance, UX, etc.)

Why This Matters

- Consumers make wrong purchase decisions
- Companies lack real-time feedback intelligence
- Product defects are discovered too late → financial loss + brand damage

Limitations of Existing Systems

- Only polarity-based sentiment (Positive/Negative)
- No attribute-level breakdown
- No competitive benchmarking
- Requires manual data science pipelines

Proposed Improvement - A system that introduces:

- Contextual Weighing NLP
- Attribute-level sentiment extraction
- Real-time benchmarking across competitors
- Automated business recommendation generation

3. Proposed Solution

High-Level Idea - A centralized AI system that ingests reviews from multiple sources and converts them into structured business intelligence dashboards.

Workflow (Step-by-Step)

1. Data Ingestion

- Load reviews from CSV / API sources
- Normalize product categories

2. NLP Processing

- Clean text (stopwords, noise removal)
- Sentiment scoring using NLP models
- Attribute extraction (battery, build, UX, etc.)

3. Sentiment Intelligence Engine

- Compute Sentix Score (0–100)
- Perform contextual weighting

- Cluster pain points

4. Visualization Layer

- Sentiment distribution charts
- Attribute-level bar graphs
- Word clouds for pain-point detection

5. Insight Generation

- Auto-generated strategic summaries
- “Verified Buy Recommendation”
- Competitive comparison insights

Key Differentiator - Unlike existing tools, Sentix Pulse acts as a:

"Virtual Product Analyst" that explains WHY a product performs well or fails - not just whether it is positive or negative."

4. System Architecture

Core Modules

1. Frontend Layer

- Streamlit-based interactive dashboard
- Glassmorphism UI design
- Role-based dashboards (Customer vs Corporate)

2. Authentication Layer

- Role-Based Access Control (RBAC)
- Secure login system with 2FA (SMTP-based email verification)

3. NLP Engine

- Sentiment analysis (TextBlob / NLTK)
- WordCloud generation
- Feature extraction & contextual scoring

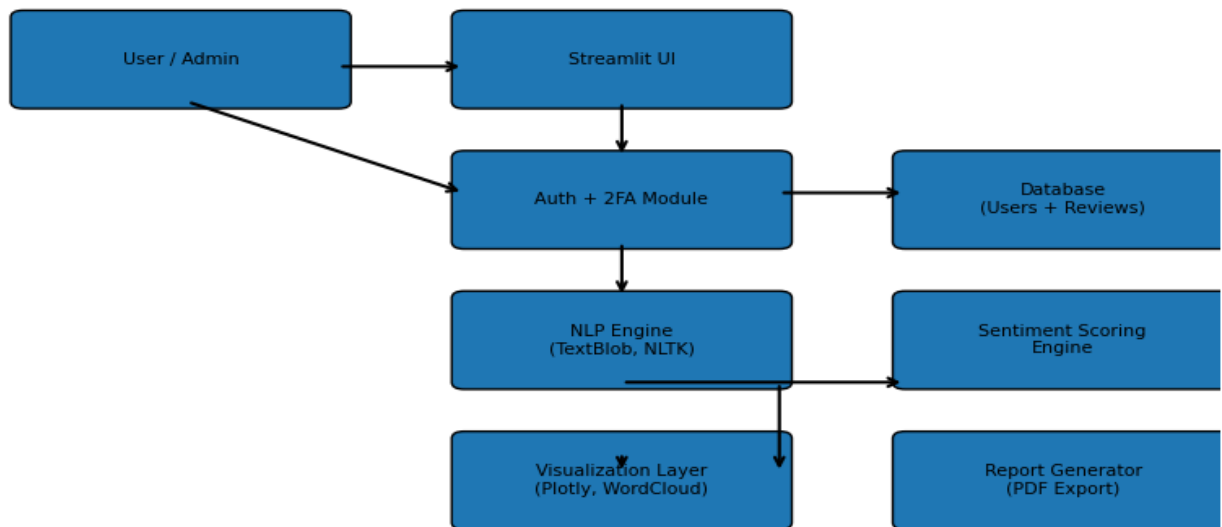
4. Analytics Engine

- Sentix Score computation
- Attribute-level aggregation

- Competitive benchmarking logic

5. Reporting Module

- PDF report generator for corporate insights
- Exportable dashboards for audits



5. Technology Stack

- **Language:** Python 3.10+
- **Frontend/UI:** Streamlit
- **Data Processing:** Pandas, NumPy
- **NLP:** NLTK, TextBlob
- **Visualization:** Plotly, Matplotlib, WordCloud
- **Security:** SMTP (TLS-based 2FA authentication)
- **Reporting:** PDF generation (ReportLab / similar library)

6. Functional Requirements

- Secure user registration with 2FA
- Role-based dashboards (Customer / Corporate)

- Product search with hierarchical filters:
 - Category → Brand → Model
- Sentiment analysis at:
 - Review level
 - Attribute level
- Generation of:
 - Sentix Score
 - Pain-point clusters
- Exportable reports (PDF)
- Access restriction based on company mapping

7. Non-Functional Requirements

- **Performance:** UI response < 500ms per interaction
- **Scalability:** Modular architecture for adding new product domains
- **Security:** Hashed credentials + SMTP verification
- **Reliability:** Graceful handling of missing or sparse data
- **UX:** High-contrast cyber-themed UI for analytical clarity

8. Data Design

Users Table

- fname
- lname
- email
- password (hashed)
- role (Customer / Corporate)
- company

Reviews Table

- category
- brand

- model
- review_text
- sentiment_label

Relationship Logic

- Corporate users are restricted via company filter
- Ensures data isolation across organizations

9. Algorithm / Logic Design

Sentiment Scoring Logic

IF review contains positive keywords:

score += 1

IF review contains negative keywords:

score -= 1

IF score > 0:

sentiment = Positive

ELSE:

sentiment = Negative

Verified Buy Score

$$V = \left(\frac{P_{positive}}{P_{total}} \right) \times 100$$

Decision Rule:

- If $V > 68\%$ → **Recommend Product**
- Else → Not Recommended / Needs Improvement

10. UI/UX Design

Design Philosophy

- Minimalist executive dashboard
- Data-first visualization approach
- No clutter, high-density insights

Key Screens

- Login Portal (Role-based toggle)
- Customer Dashboard
- Corporate Analytics Dashboard

Visual Components

- Sentix Score Gauge (primary focus)
- Attribute comparison bar charts
- Word cloud for pain-point detection
- Competitive benchmarking panels

11. Implementation Plan

- **Phase 1:** Data structure + environment setup
- **Phase 2:** NLP sentiment engine development
- **Phase 3:** Authentication + role-based filtering
- **Phase 4:** Dashboard + visualization layer
- **Phase 5:** Testing + deployment

12. Testing Strategy

- **Unit testing:**
 - Authentication flow
 - SMTP verification
- **Integration testing:**
 - Role-based access control

- **Edge case testing:**
 - Products with zero reviews
 - Empty or noisy datasets
- **Security testing:**
 - Unauthorized access prevention

13. Risks & Limitations

- **Sarcasm Detection Weakness**
 - Mitigation: Contextual keyword weighting
- **Sparse Data for New Products**
 - Mitigation: “Insufficient Data” fallback system
- **Basic NLP Accuracy Limits**
 - Future upgrade required for transformer models

14. Future Enhancements

- Real-time review scraping (Selenium / APIs)
- Upgrade to transformer-based NLP (BERT / GPT APIs)
- Multi-product comparison engine
- Predictive sentiment forecasting
- Market trend detection dashboard

15. Deployment Plan

- **Local Execution:** streamlit run app.py
- **Cloud Deployment:**
 - Streamlit Community Cloud (basic)
 - AWS EC2 (enterprise-grade deployment)
- **CI/CD Pipeline:**
 - GitHub Actions for automated deployment
 - Version-controlled model updates

